MEDICONNECT HOSPITAL MANAGEMENT SYSTEM

Portfolio Case Study

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SELF-DIRECTED PROJECT SIMULATION , NOT A REAL HOSPITAL IMPLEMENTATION

Project at a glance

Field	Details
Project	MediConnect – Hospital Management System
Type	Self-directed project simulation
Focus	Project coordination and business analysis
Proposed users	Hospital management, receptionists, doctors, nurses, billing staff and patients
Proposed solution	A web-based MVP for patient registration, appointments, medical records, billing and management reporting
My practice role	Project Coordinator / Business Analyst
Implementation status	Not built or deployed; portfolio documents and planning artefacts only

1. Background and problem

MediConnect was created as a practical case study around a hospital assumed to rely on paper records and manual administrative processes. The scenario includes delays in patient registration, difficulty locating medical files, manual appointment handling, billing-entry errors and limited information sharing between departments. These are scenario assumptions used for the exercise, not findings from a real hospital assessment.

2. Project objective

The proposed objective was to plan an integrated web-based system that could make core hospital administrative and information workflows more organised, while ensuring that access to patient information is restricted to authorised users.

3. Proposed MVP scope

- Patient registration, profiles, unique patient IDs, search and duplicate-record warnings.
- Appointment creation, rescheduling, cancellation and status tracking.
- Electronic medical-record retrieval and authorised clinical updates.
- Billing, automatic bill totals, payment records and outstanding balances.
- Role-based accounts and controlled information sharing across departments.
- Management dashboard and agreed operational reports.

Features outside the initial proposed release included telemedicine, ambulance tracking, advanced pharmacy inventory, full laboratory integration, insurance claims automation, a dedicated mobile application and external integrations not agreed during initial requirements work.

4. My role and activities

For this learning project, I worked through the responsibilities expected of a Project Coordinator with supporting Business Analyst activities. The work focused on organising project information, translating the scenario into requirements, planning activities and practising progress and issue coordination.

- Outlined the project purpose, objectives, proposed scope, stakeholders, assumptions, constraints and acceptance criteria in a project charter.
- Prepared a draft Business Requirements Document with 21 functional requirements and initial non-functional requirements.
- Used MoSCoW prioritisation to distinguish essential requirements from lower-priority or future-phase requests.
- Created a Work Breakdown Structure to organise initiation, requirements, design, development, testing, UAT, deployment, handover and closure.
- Prepared an Excel Gantt chart and considered task dependencies, parallel work and schedule risks.
- Created weekly status reporting templates to track planned versus actual progress, blockers, risks, actions and next-period priorities.
- Practised risk identification, issue escalation, change-request assessment, stakeholder communication and action tracking.
- Prepared UAT, deployment, training and handover templates to define readiness and acceptance evidence.

5. Requirements and analysis highlights

The requirements work translated broad operational problems into system behaviours. Examples included unique patient identifiers, duplicate-record warnings, role-based access, audit history for clinical corrections, billing calculations and controlled emergency access. The exercise also highlighted areas requiring stakeholder confirmation, including patient portal permissions, billing approval rules, performance targets, integrations and data migration.

6. Planning and coordination approach

- The WBS was used to break the work into phases and manageable activities.
- The Gantt chart was used to represent estimated durations and dependencies.
- Progress reporting compared planned and actual percentages in an illustrative weekly snapshot.
- Blockers were assigned to appropriate technical or business owners, with escalation and schedule impact reviewed by the Project Manager.
- Change requests were not treated as automatic additions to scope; their value, effort, cost, schedule and risk were to be assessed before approval.

Planning review finding: the project charter states a three-month target, while the detailed Gantt chart represents approximately 21 calendar weeks. This discrepancy is recorded as an item requiring schedule reconciliation rather than being hidden.

7. Risk and issue management

The practice risk register considered data migration, staff resistance, unstable internet, data quality, scope changes, information security, integration and resource/schedule constraints.

The issue log used simulated examples such as missing records after migration, duplicate or incomplete records and unstable connectivity. The response approach emphasised investigation, containment, assigning owners, escalation, corrective action and verification before proceeding.

8. Testing, deployment and handover

A UAT plan was prepared with sample scenarios for registration, patient search, duplicate warnings, appointments, clinical notes, role permissions, billing, reporting, audit history and connectivity interruption. Actual results were intentionally left blank because no test execution occurred.

The deployment plan proposed a controlled phased rollout, readiness checks, role-based staff training, support communications and handover of documentation, access processes, backup/recovery procedures and open risks. No real deployment, training or handover is claimed.

9. Portfolio artefacts

Artefact	Purpose
Project Charter	Defines the business need, objectives, scope, resources, milestones and acceptance criteria.
Business Requirements Document	Records the proposed business and system requirements.
Work Breakdown Structure	Breaks the project into phases, activities and responsible roles.
Gantt Chart	Shows the proposed schedule, durations and dependencies.
Weekly Status Report (Word and Excel)	Demonstrates progress reporting, blockers, risks and action tracking.
Risk Register and Issue Log	Records potential risks and simulated active issues with responses and owners.
UAT Plan and Sign-off Template	Defines sample acceptance tests and the approval record.
Deployment, Training and Handover Plan	Defines proposed rollout, user preparation and operational handover.
Project Closure Report	Records the simulation's closure position, limitations and lessons learned.

10. What I learned

- Good coordination involves following up actions and dependencies, not only arranging meetings.
- Requirements need to be clear, testable and confirmed by the people who will use the system.
- Project plans are working documents and should be updated when assumptions, resources or dependencies change.
- Risks need proactive monitoring, while issues need timely investigation and escalation.
- Scope changes should be assessed and approved before the team commits to them.
- Project completion requires evidence of testing, acceptance, training and handover.

11. Limitations and next steps

This is a documentation-led simulation. There was no real client, hospital discovery, development team, software build, live data, production deployment or formal sponsor acceptance. The sample progress figures and issue examples are illustrative. The next step for a real project would be to validate the assumptions with hospital stakeholders, confirm scope and schedule, and then proceed through delivery with recorded evidence.